Discovering the Signal: It Lives in the Tail

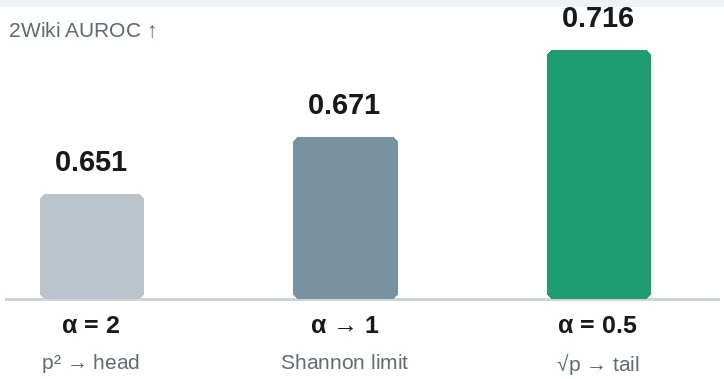
42-signal survey → entropy decomposition

1 Broad survey

42 confidence signals · 4 datasets × 1,000 questions

$$H_\alpha(p) = \frac{1}{(1-\alpha)} \cdot \log \sum_v p(v)^\alpha$$

α > 1: head-heavy · α → 1: Shannon · α < 1: tail-sensitive



Lower α gives rare tokens more influence → better separation

2 Decompose entropy

A = top-5, B = tail, $m_5 = P(A)$

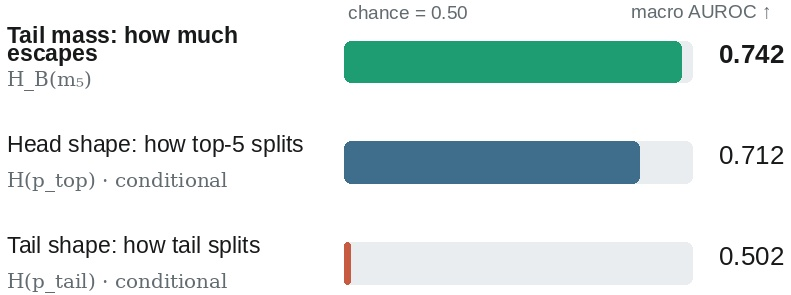
$p_{top}(v) = p(v)/m_5$ on A; $p_{tail}(v) = p(v)/(1-m_5)$ on B

$H(p) = -\sum_{v \in A} p(v) \log p(v) - \sum_{v \in B} p(v) \log p(v)$

Substitute $p = m_5 p_{top}$ on A and $p = (1-m_5) p_{tail}$ on B; expand log

$$\Rightarrow H = H_B(m_5) + m_5 H(p_{top}) + (1-m_5) H(p_{tail})$$

Diagnostic probes isolate shape: H(p_tail) normalizes away tail amount



Amount matters (.742); allocation inside the tail does not (.502)

Takeaway: tail-sensitive signals detect how much mass escapes the head—not how that tail mass is arranged.

From Entropy to tl_sum

isolate the escape signal → make it monotonic → aggregate across tokens

1 Identify the strongest term

$$H_B(m_5)$$

escape 0.742 > full entropy 0.726

discard within-top-5 and tail-composition terms

→

2 Design a monotonic score

$$H_B(m_5) \rightarrow -\log m_5^{(t)}$$

H_B is not globally monotonic

less top-5 mass → larger score

→

3 Aggregate over the answer

$$tl_sum = \sum_t -\log m_5^{(t)} = -\log \prod_t m_5^{(t)}$$

joint surprisal of staying inside top-5

Why sum?

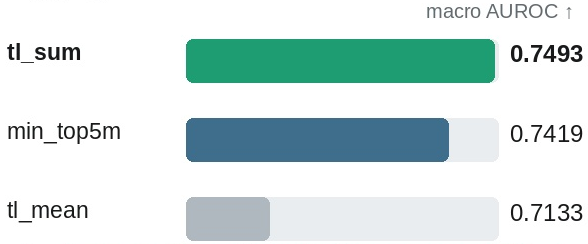
wrong answers often leak a little at several positions

token t_1 — small leak + token t_2 — small leak + token t_3 — small leak → Σ tl_sum

Worst token: keeps one position

tl_sum: all positions accumulate

Aggregation scan



tl_sum vs. min_top5m: paired bootstrap p < 0.001

Takeaway: tl_sum is built from the top-5 escape signal identified by entropy decomposition—not an ad hoc formula.

Why Top-5? Deterministic, Not Random

At each decoding step, rank the full-vocabulary softmax probabilities.

What happens at token t?

deterministic at every step

logits at t — one score / token → **softmax** — p_t(v) over |V| → **sort ↓** — largest to smallest

Illustrative ranked distribution



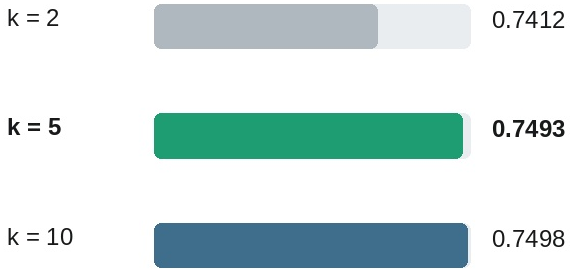
$$m_t = \sum_{i=1}^{5} p_{(i)}^{(t)} \qquad \text{tail mass} = 1 - m_t$$

No sampling: A_t is always the five largest probabilities under p_t.

Tail means ranks 6…|V|—not the bottom five tokens.

Why choose k = 5?

tl_sum@k = Σ_t −log m_{k,t} · macro AUROC



k=5 ≈ k=10 (p=.824); k=2 is lower (p=.022)

- Allows several casing, alias, and subword variants.
- Still keeps the candidate set narrow.

Post-hoc sensitivity on the same frozen 4×1,000 direct traces.

Takeaway: top-5 is deterministic; k=5 is a validated but model- and tokenizer-dependent hyperparameter.

Entropy Decomposition: Step-by-Step

Split the vocabulary into the top-5 set A and the tail B; drop token index t for readability.

$A = V_5(p), \quad B = V \setminus A \qquad m = \sum_{v \in A} p(v) \qquad p_{top}(v) = p(v)/m$ for v in A $\qquad p_{tail}(v) = p(v)/(1-m)$ for v in B

Algebraic identity

exact—not an approximation

1. $H(p) = -\sum_{v \in A} p(v) \log p(v) \; - \; \sum_{v \in B} p(v) \log p(v)$

2. $= -\sum_{v \in A} m\, p_{top}(v) \log[m\, p_{top}(v)] \; -\sum_{v \in B} (1-m)\, p_{tail}(v) \log[(1-m)\, p_{tail}(v)]$

3. $= -m \log m - (1-m) \log(1-m) + m\, H(p_{top}) + (1-m)\, H(p_{tail})$

$$H(p) = H_B(m) + m\, H(p_{top}) + (1-m)\, H(p_{tail})$$

Empirical probes (conditional; approximate)

The exact identity weights the last two by m and 1−m.

$H_B(m)$ — top-5 vs. tail membership — 0.742

$H(p_{top})$ — competition inside top-5 — 0.712

$H(p_{tail})$ — composition inside the tail — 0.502

Algebraic decomposition ends here.

Design: $TL_t = -\log m_t$; $tl_sum = \sum_t TL_t$

Takeaway: decomposition locates the escape signal; surprisal is a motivated scoring choice, not an equality.